

Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

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1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

1.2 Задачи лабораторной работы

- 1 Выполнить приимеры
- 2 Выполнить дествия по работе с каталогами и файлами
- 3 Выполнить действия с правами доступа
- 4 Получить дополнительные сведения при помощи справки по командам.

2. 2. Процесс выполнения лабораторной работы

2.1 Выполнение примеров

```
iaatabaeva@iaatabaeva:~$ touch abc1
iaatabaeva@iaatabaeva:~$ cp abc1 april
iaatabaeva@iaatabaeva:~$ cp abc1 may
iaatabaeva@iaatabaeva:~$ mkdir monthly
iaatabaeva@iaatabaeva:~$ cp april may monthly
iaatabaeva@iaatabaeva:~$ cp monthly/may monthly/june
iaatabaeva@iaatabaeva:~$ ls monthly
april  june  may
iaatabaeva@iaatabaeva:~$ mkdir monthly.00
iaatabaeva@iaatabaeva:~$ cp -r monthly monthly.00
iaatabaeva@iaatabaeva:~$ cp -r monthly.00 /tmp
iaatabaeva@iaatabaeva:~$
```

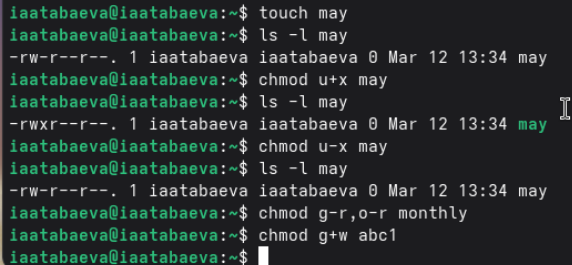
Рисунок 1: Выполнение примеров

2.2 Выполнение примеров

```
iaatabaeva@iaatabaeva:~$  
iaatabaeva@iaatabaeva:~$ mv april july  
iaatabaeva@iaatabaeva:~$ mv july monthly.00  
iaatabaeva@iaatabaeva:~$ ls monthly.00  
july  monthly  
iaatabaeva@iaatabaeva:~$ mv monthly.00 monthly.01  
iaatabaeva@iaatabaeva:~$ mkdir reports  
iaatabaeva@iaatabaeva:~$ mv monthly.01 reports  
iaatabaeva@iaatabaeva:~$ mv reports/monthly.01 reports/monthly  
iaatabaeva@iaatabaeva:~$
```

Рисунок 2: Выполнение примеров

2.3 Выполнение примеров

A terminal window with a dark background and green text. It shows a series of commands and their outputs. The user 'iaatabaeva' is in the home directory. The commands are: 'touch may', 'ls -l may', 'chmod u+x may', 'ls -l may', 'chmod u-x may', 'ls -l may', 'chmod g-r,o-r monthly', and 'chmod g+w abc1'. The output of 'ls -l may' shows the file 'may' with permissions '-rw-r--r--', owned by 'iaatabaeva', with a size of 0, created on 'Mar 12 13:34'. The word 'may' is highlighted in green in the output. A cursor is visible at the end of the last command line.

```
iaatabaeva@iaatabaeva:~$ touch may
iaatabaeva@iaatabaeva:~$ ls -l may
-rw-r--r--. 1 iaatabaeva iaatabaeva 0 Mar 12 13:34 may
iaatabaeva@iaatabaeva:~$ chmod u+x may
iaatabaeva@iaatabaeva:~$ ls -l may
-rwxr--r--. 1 iaatabaeva iaatabaeva 0 Mar 12 13:34 may
iaatabaeva@iaatabaeva:~$ chmod u-x may
iaatabaeva@iaatabaeva:~$ ls -l may
-rw-r--r--. 1 iaatabaeva iaatabaeva 0 Mar 12 13:34 may
iaatabaeva@iaatabaeva:~$ chmod g-r,o-r monthly
iaatabaeva@iaatabaeva:~$ chmod g+w abc1
iaatabaeva@iaatabaeva:~$
```

Рисунок 3: Выполнение примеров

2.4 Создание директорий и копирование файлов

```
iaatabaeva@iaatabaeva:~$  
iaatabaeva@iaatabaeva:~$ cp /usr/include/linux/sysinfo.h ~  
iaatabaeva@iaatabaeva:~$ mv sysinfo.h equipment  
iaatabaeva@iaatabaeva:~$ mkdir ski.plases  
iaatabaeva@iaatabaeva:~$ mv equipment ski.plases/  
iaatabaeva@iaatabaeva:~$ mv ski.plases/equipment ski.plases/equiplist  
iaatabaeva@iaatabaeva:~$ touch abc1  
iaatabaeva@iaatabaeva:~$ cp abc1 ski.plases/equiplist2  
iaatabaeva@iaatabaeva:~$ cd ski.plases/  
iaatabaeva@iaatabaeva:~/ski.plases$ mkdir equipment  
iaatabaeva@iaatabaeva:~/ski.plases$ mv equiplist equipment/  
iaatabaeva@iaatabaeva:~/ski.plases$ mv equiplist2 equipment/  
iaatabaeva@iaatabaeva:~/ski.plases$ mkdir newdir  
iaatabaeva@iaatabaeva:~/ski.plases$ mv newdir ski.plases/  
iaatabaeva@iaatabaeva:~/ski.plases$ mv ski.plases/newdir/ ski.plases/plans  
mv: cannot stat 'ski.plases/newdir/': No such file or directory  
iaatabaeva@iaatabaeva:~/ski.plases$
```

Рисунок 4: Работа с каталогами

2.5 Работа с командой chmod

```
iaatabaeva@iaatabaeva:~/ski.plases$  
iaatabaeva@iaatabaeva:~/ski.plases$ mkdir australia play  
iaatabaeva@iaatabaeva:~/ski.plases$ touch my_os feathers  
iaatabaeva@iaatabaeva:~/ski.plases$ chmod 744 australia/  
iaatabaeva@iaatabaeva:~/ski.plases$ chmod 711 play/  
iaatabaeva@iaatabaeva:~/ski.plases$ chmod 544 my_os  
iaatabaeva@iaatabaeva:~/ski.plases$ chmod 664 feathers  
iaatabaeva@iaatabaeva:~/ski.plases$ ls -l  
total 0  
drwxr--r--. 1 iaatabaeva iaatabaeva  0 Mar 12 13:36 australia  
drwxr-xr-x. 1 iaatabaeva iaatabaeva 38 Mar 12 13:35 equipment  
-rw-rw-r--. 1 iaatabaeva iaatabaeva  0 Mar 12 13:36 feathers  
-r-xr--r--. 1 iaatabaeva iaatabaeva  0 Mar 12 13:36 my_os  
drwx--x--x. 1 iaatabaeva iaatabaeva  0 Mar 12 13:36 play  
drwxr-xr-x. 1 iaatabaeva iaatabaeva  0 Mar 12 13:35 ski.plases  
iaatabaeva@iaatabaeva:~/ski.plases$
```

Рисунок 5: Настройка прав доступа

2.6 Файл /etc/passwd

```
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/:/usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:/usr/sbin/nologin
polkitd:x:114:114:User for polkitd:/:/sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
chrony:x:996:996:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
unbound:x:993:993:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
clevis:x:992:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:/usr/sbin/nologin
gluster:x:991:991:GlusterFS daemons:/run/gluster:/usr/sbin/nologin
qemu:x:107:107:qemu user:/:/usr/sbin/nologin
/etc/passwd
```

Рисунок 6: Файл /etc/passwd

2.7 Работа с файлами и правами доступа

```
iaatabaeva@iaatabaeva:~$ touch my_os feathers
iaatabaeva@iaatabaeva:~$ cp feathers file.old
iaatabaeva@iaatabaeva:~$ mv file.old play/
mv: cannot move 'file.old' to 'play/': Not a directory
iaatabaeva@iaatabaeva:~$ mkdir australia play
iaatabaeva@iaatabaeva:~$ mv file.old play/
iaatabaeva@iaatabaeva:~$ mkdir fun
iaatabaeva@iaatabaeva:~$ cp -R play/ fun/
iaatabaeva@iaatabaeva:~$ mv fun/ play/games
iaatabaeva@iaatabaeva:~$ chmod u-r feathers
iaatabaeva@iaatabaeva:~$ cat feathers
cat: feathers: Permission denied
iaatabaeva@iaatabaeva:~$ cp feathers feathers2
cp: cannot open 'feathers' for reading: Permission denied
iaatabaeva@iaatabaeva:~$ chmod u+r feathers
iaatabaeva@iaatabaeva:~$ chmod u-x play/
iaatabaeva@iaatabaeva:~$ cd play/
bash: cd: play/: Permission denied
iaatabaeva@iaatabaeva:~$ chmod +x play/
iaatabaeva@iaatabaeva:~$
```

2.8 Справка по командам

```
MOUNT(8)                                System Administration                                MOUNT(8)

NAME
    mount - mount a filesystem

SYNOPSIS
    mount [-h|-V]

    mount [-l] [-t fstype]

    mount -a [-fFnrsvw] [-t fstype] [-O optlist]

    mount [-fnrsvw] [-o options] device|mountpoint

    mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

    mount --bind|--rbind|--move olddir newdir

    mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable] mountpoint

DESCRIPTION
    All files accessible in a Unix system are arranged in one big tree, the file hierarchy, rooted at /.
    These files can be spread out over several devices. The mount command serves to attach the filesystem
    found on some device to the big file tree. Conversely, the umount(8) command will detach it again. The
    filesystem is used to control how data is stored on the device or provided in a virtual way by network
    or other services.

    The standard form of the mount command is:

        mount -t type device dir

    This tells the kernel to attach the filesystem found on device (which is of type type) at the
    directory dir. The option -t type is optional. The mount command is usually able to detect a
    filesystem. The root permissions are necessary to mount a filesystem by default. See section

Manual page mount(8) line 1 (press h for help or q to quit)
```

2.9 Справка по командам

```
FSCK(8)                                     System Administration                                     FSCK(8)

NAME
    fsck - check and repair a Linux filesystem

SYNOPSIS
    fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--] [fs-specific-options]

DESCRIPTION
    fsck is used to check and optionally repair one or more Linux filesystems. filesystem can be a device
    name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a filesystem label or UUID
    specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or LABEL=root). Normally, the fsck program
    will try to handle filesystems on different physical disk drives in parallel to reduce the total
    amount of time needed to check all of them.

    If no filesystems are specified on the command line, and the -A option is not specified, fsck will
    default to checking filesystems in /etc/fstab serially. This is equivalent to the -As options.

    The exit status returned by fsck is the sum of the following conditions:

    0
        No errors

    1
        Filesystem errors corrected

    2
        System should be rebooted

    4
        Filesystem errors left uncorrected

    8
        Operational error

Manual page fsck(8) line 1 (press h for help or q to quit)
```

2.10 Справка по командам

```

MKFS(8)                                     System Administration                               MKFS(8)

NAME
    mkfs - build a Linux filesystem

SYNOPSIS
    mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
    This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

    mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The device
    argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file that shall contain
    the filesystem. The size argument is the number of blocks to be used for the filesystem.

    The exit status returned by mkfs is 0 on success and 1 on failure.

    In actuality, mkfs is simply a front-end for the various filesystem builders (mkfs.fstype) available
    under Linux. The filesystem-specific builder is searched for via your PATH environment setting only.
    Please see the filesystem-specific builder manual pages for further details.

OPTIONS
    -t, --type type
        Specify the type of filesystem to be built. If not specified, the default filesystem type
        (currently ext2) is used.

    fs-options
        Filesystem-specific options to be passed to the real filesystem builder.

    -V, --verbose
        Produce verbose output, including all filesystem-specific commands that are executed. Specifying
        this option more than once inhibits execution of any filesystem-specific commands. This is really
        only useful for testing.

Manual page mkfs(8) line 1 (press h for help or q to quit)
```

Рисунок 10: Команда mkfs
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2.11 Справка по командам

```
KILL(1)                                     User Commands                               KILL(1)

NAME
    kill - terminate a process

SYNOPSIS
    kill [-signal|-s signal|-p] [-q value] [-a] [--timeout milliseconds signal] [--] pid|name...

    kill -l [number|0xsigmask] | -L

    kill -d pid

DESCRIPTION
    The command kill sends the specified signal to the specified processes or process groups.

    If no signal is specified, the TERM signal is sent. The default action for this signal is to terminate the process. This signal should be used in preference to the KILL signal (number 9), since a process may install a handler for the TERM signal in order to perform clean-up steps before terminating in an orderly fashion. If a process does not terminate after a TERM signal has been sent, then the KILL signal may be used; be aware that the latter signal cannot be caught, and so does not give the target process the opportunity to perform any clean-up before terminating.

    Most modern shells have a builtin kill command, with a usage rather similar to that of the command described here. The --all, --pid, and --queue options, and the possibility to specify processes by command name, are local extensions.

    If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS
    The list of processes to be signaled can be a mixture of names and PIDs.

    pid
        Each pid can be expressed in one of the following ways:
```

Manual page kill(1) line 1 (press h for help or q to quit)

3. 3. Выводы по проделанной работе

В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.